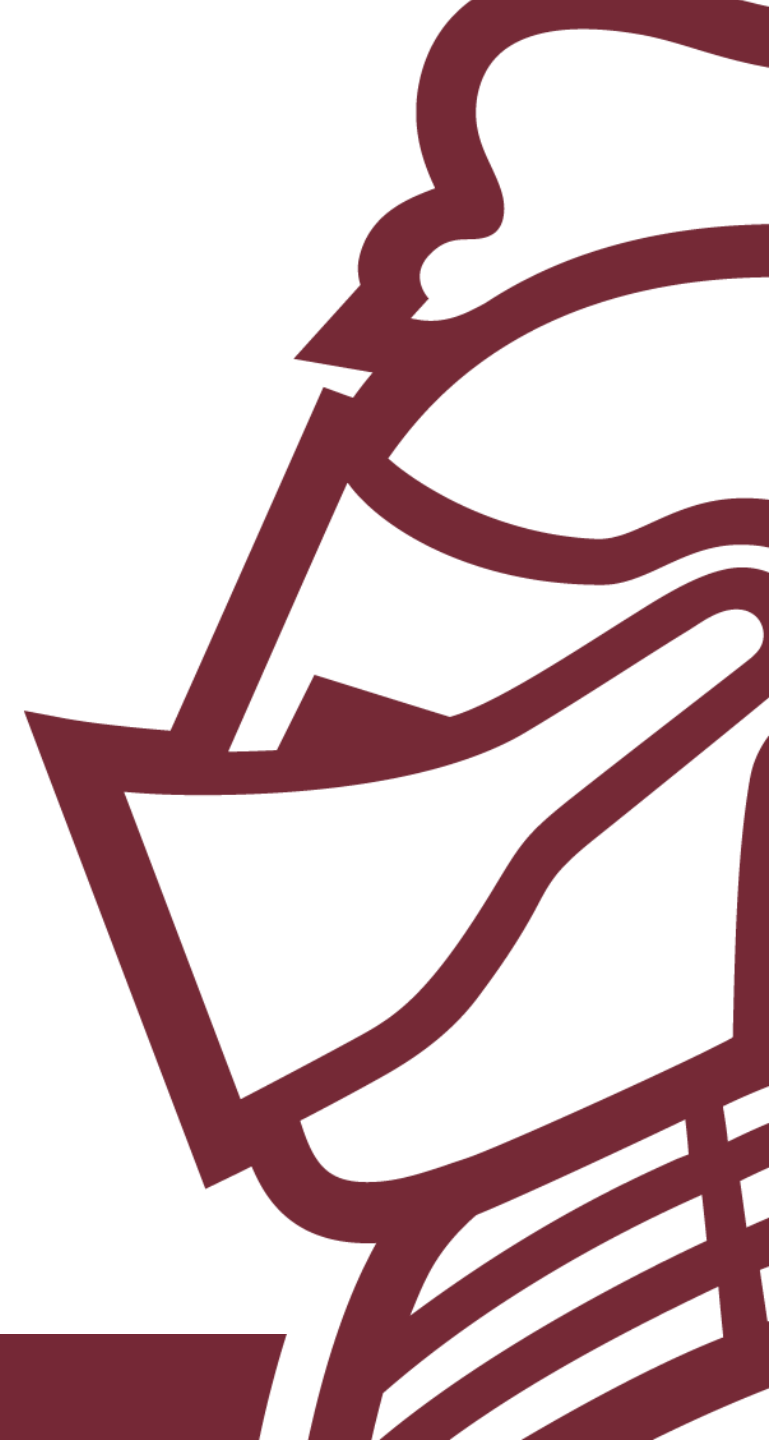


SLCM Emergency Assistance & Housing Crisis: EDA

Wiston Htoo & Nikola Mrdak
CS-430: Machine Learning





South Louisville Community Ministries (SLCM)

- Founded by churches in Louisville aiming to help those in need
- Provides immediate food, housing, and other essential needs

Problem: Too many people need SLCM resources for how much there is readily available.

Goal: Identify drivers of instability so that those who need assistance most urgently receive it.



Two Datasets

Intakes:

- 639 rows and 93 columns
- One observation = One ticket
- Contained demographic, situational, Jira, and economic data
- Missing data...

Household:

- 993 rows and 8 columns
- One observation = One Household Member
- Core Features: Number of family members, age, and family relation



Column Triage

- Keep: Actionable Case Data (Income, Housing Status)
- Optional: Messy Object dytypes (Status, Resolution, Labels 1-6)
- Drop: System Metadata (Jira IDs, Watchers, Timestamps)

Thought Process:

- Preserve useful information while trimming incomplete and vague variables



Data Quality



Missing Data: >30% missing 'Employment Status'



Inconsistent Formats: Free-text in numeric Income columns



System Quirks: Non-sequential Ticket IDs



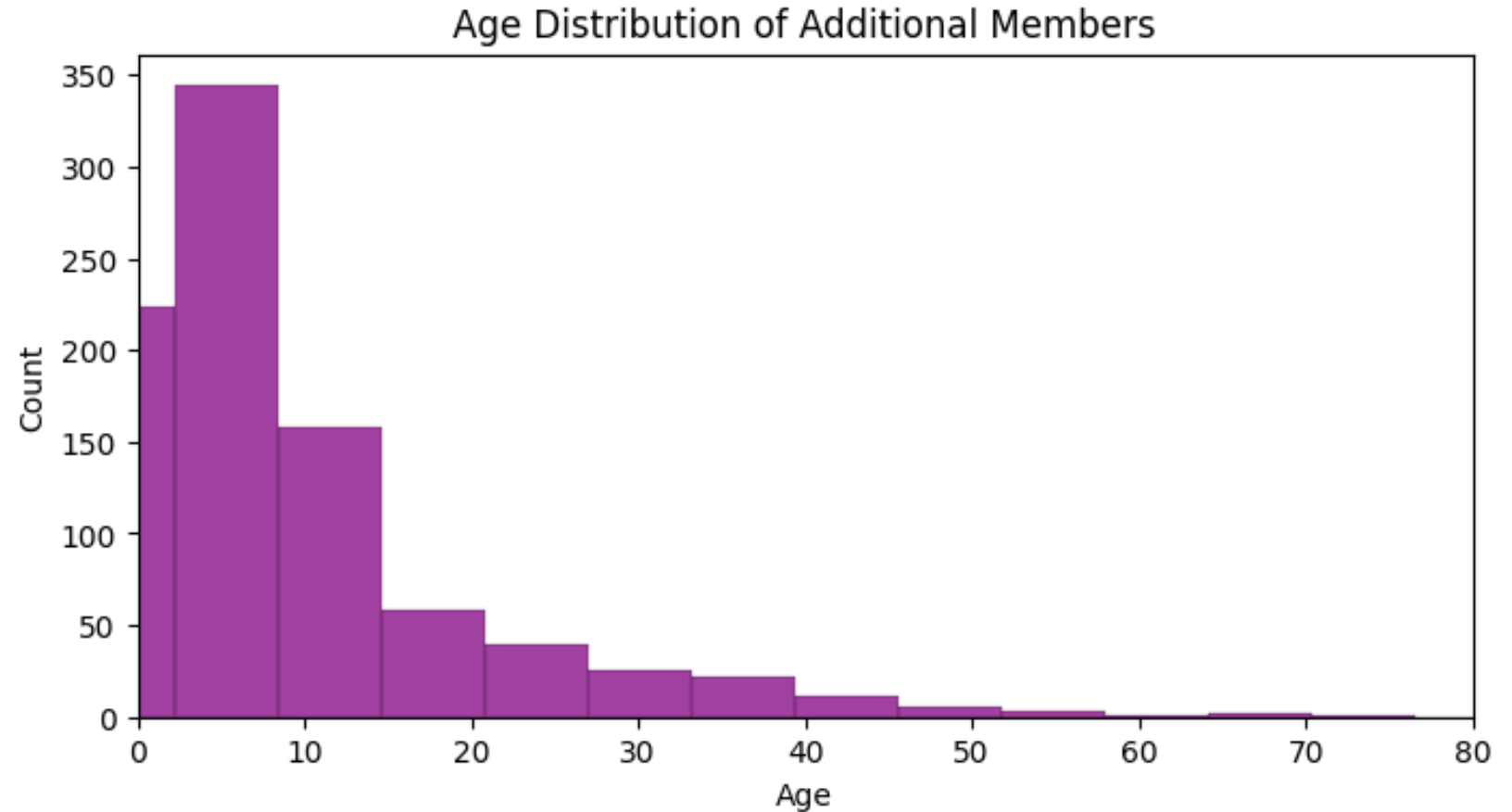
Data Preparation

- Changing dtypes for income to strict numeric format
- Filled missing Eviction Status as 'Unknown'
- For a majority of kept columns, we felt that it was best to not input missing values
- Columns with a high number of missing values and inconsistencies were prone to being dropped
- Noticed outliers in numeric variables like income.

Exploratory Data Analysis



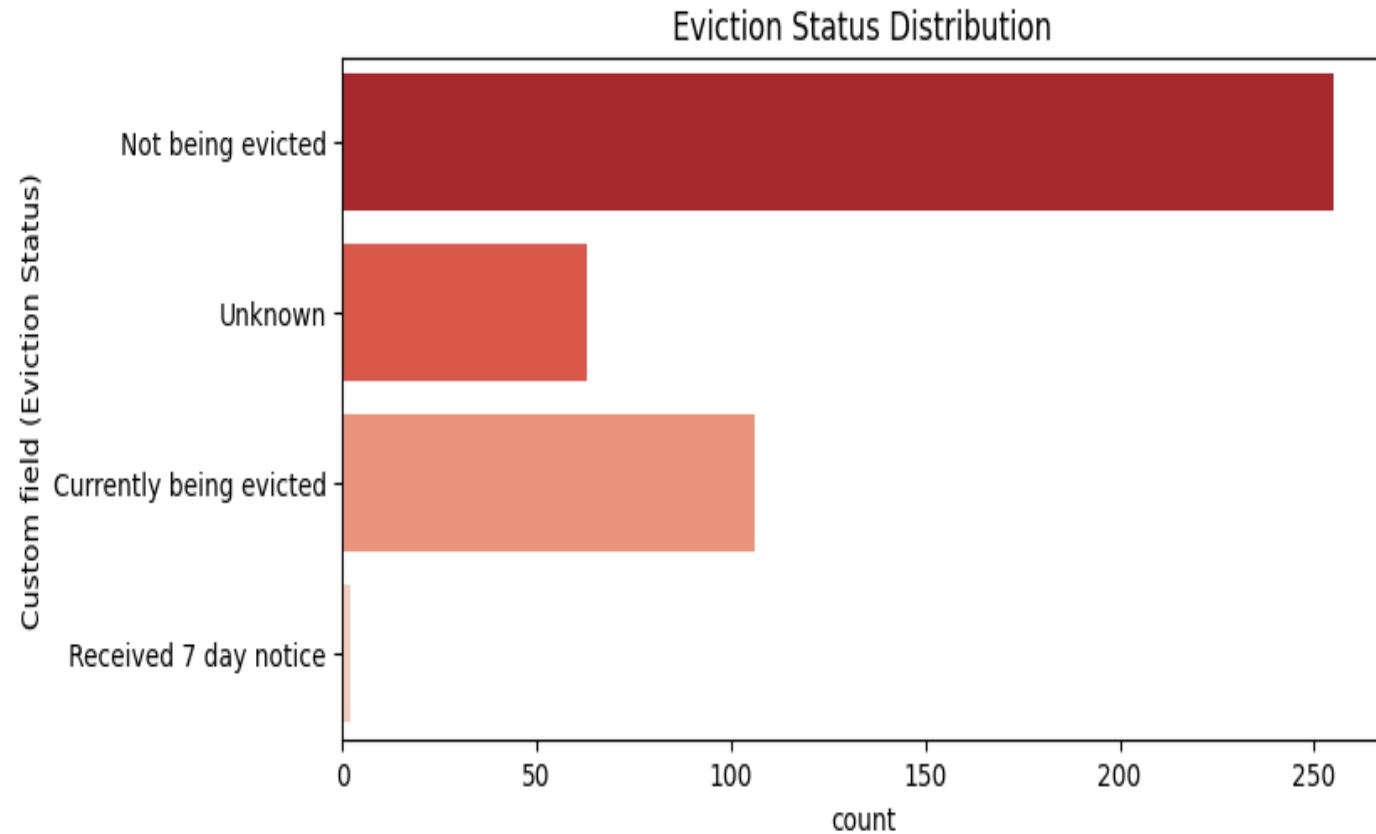
- Average Number of Additional Household Member: 2.66
- Median age of additional members: 6.0 years



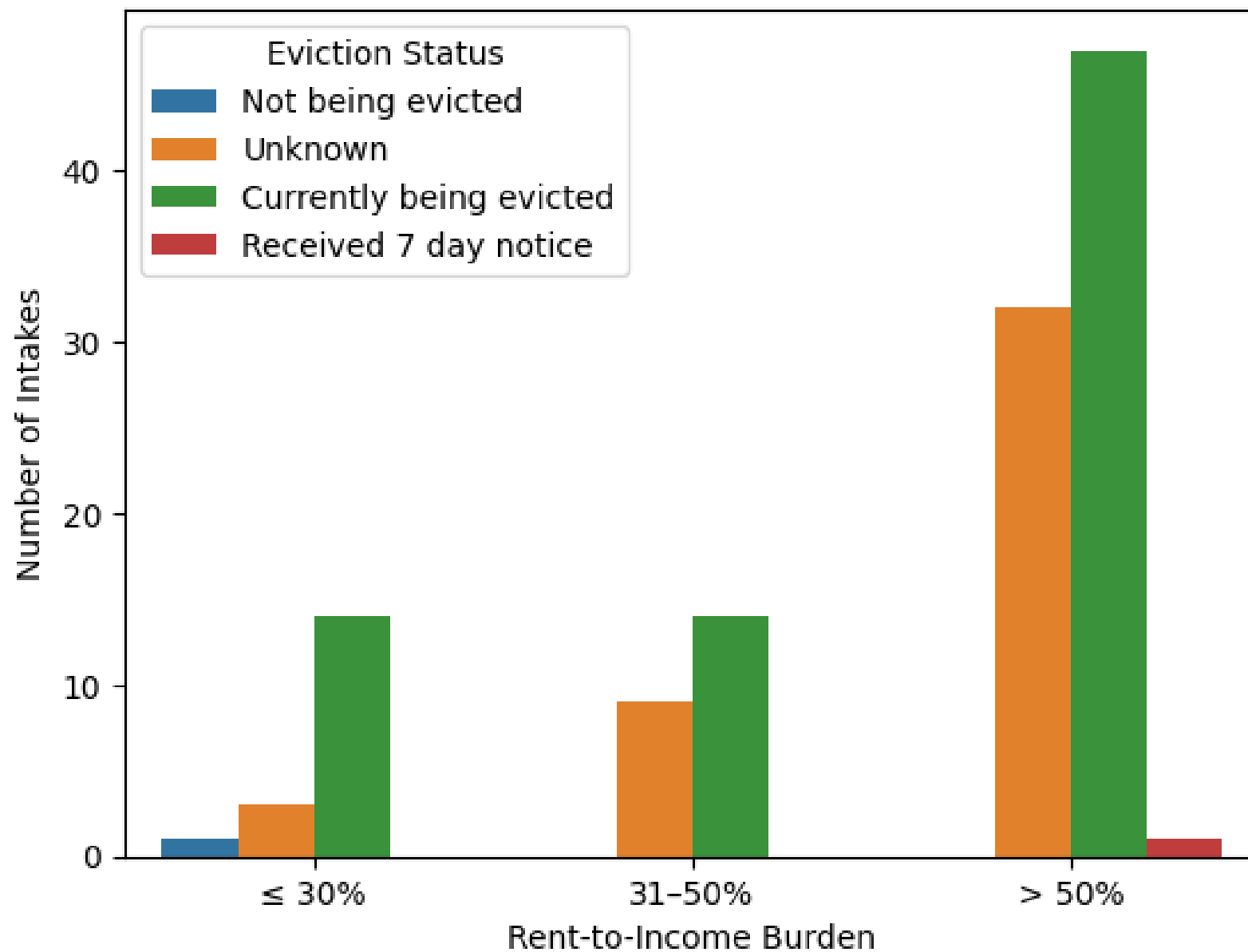
Our Research Question



Does employment status shield families from severe financial hardship and active eviction risk?



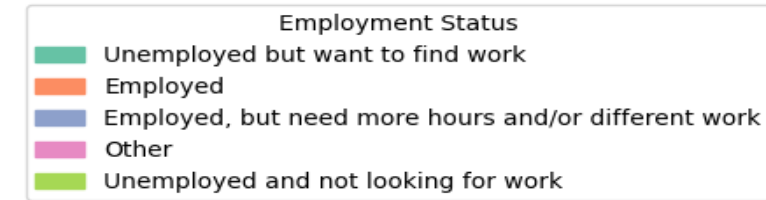
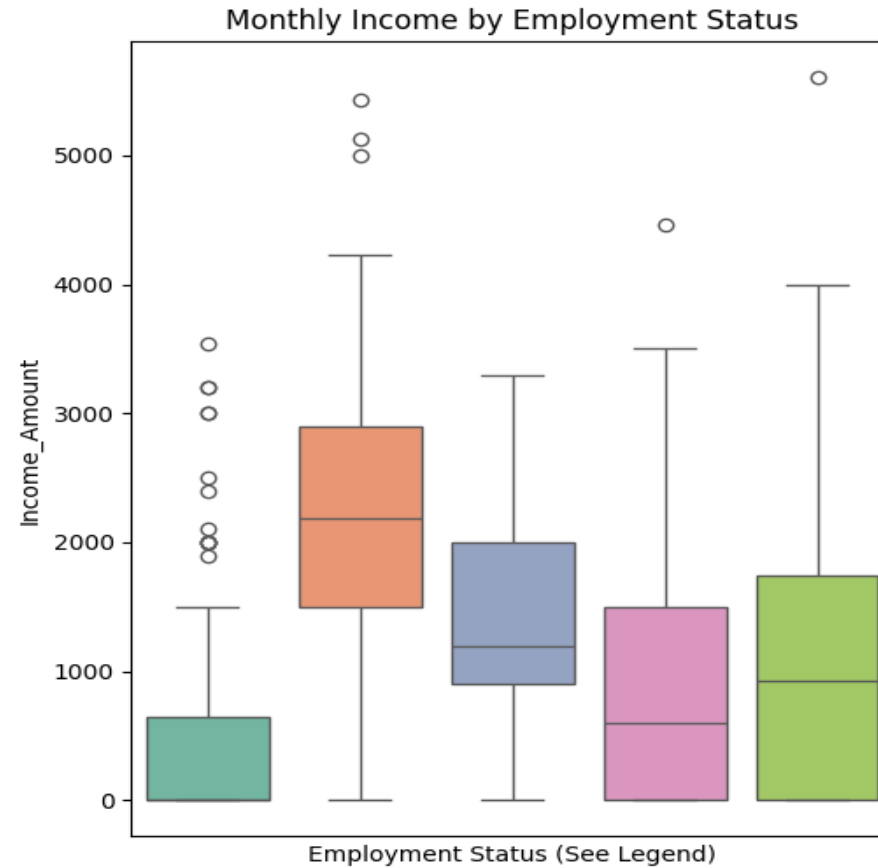
Eviction Status by Rent Burden





Key Findings So Far

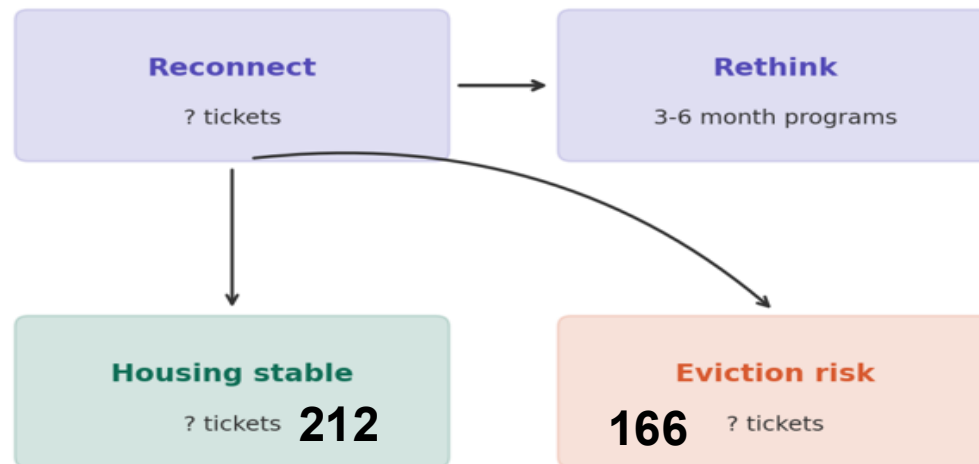
- Employment $\neq$ Financial Security
- Median Employed Income: \$2,293.50/month
- Primary Driver: Wage-to-Cost-of-Living Mismatch





Next Steps & Limitations

- Data Limitation: Captures 'Reconnect', missing 'Rebuild'
- Proposal: Implement 6-12 month follow-up metrics
- Future Modeling: Predict repeat eviction risks



Thank You

